November Top-N Strategy - Portfolio Performance: CAGR, Fund Utilisation, Sharpe and more

Rs 10,00,000 account trading the walk-forward Top 1 / 3 / 5 / 10 stocks of the 30 November dates, test years 2013-2025 | stocks and rules are always chosen from EARLIER years only | Nifty 50 = benchmark | prepared 23 Oct 2026

Question	Answer
CAGR?	**1.0% a year** for the all-dates Top 3 account (Top 1 1.0%, Top 10 0.6%) if idle cash earns nothing, **7.4%** if idle cash sits in a 6.5% liquid fund. Nifty 50 over the same years: **12.1%**. The strategy does not beat buy-and-hold.
Fund utilised?	Only **3.7% of the account** is deployed on average (positions are open on 13% of days, almost all between late October and mid-December). The trades earn about **27% a year on the capital while it is working**; the rest of the account sits idle.
Sharpe?	**-3.09** with idle cash at 0% (the account earns less than the 6.5% risk-free rate). With idle cash earning 6.5%: **0.43**, level with the Nifty (0.40) - but only because volatility is tiny: the account is just 0.9 points a year above the risk-free rate, the Nifty 5.6.
Drawdown?	Max drawdown -4.0% for Top 3 (-6.4% for Top 1) - little capital is at risk at any time; Nifty 50 maximum drawdown in the same years: -38%.
Best dates?	Choosing dates with hindsight lifts Top 3 CAGR to 2.0% (biased). Choosing them from earlier years only (2018-2025) gives 1.0% vs 1.0% for trading all dates: no real gain. Bootstrap range of Top 3 CAGR: 0.4% to 1.6% - small but reliably positive; 13 test years only.

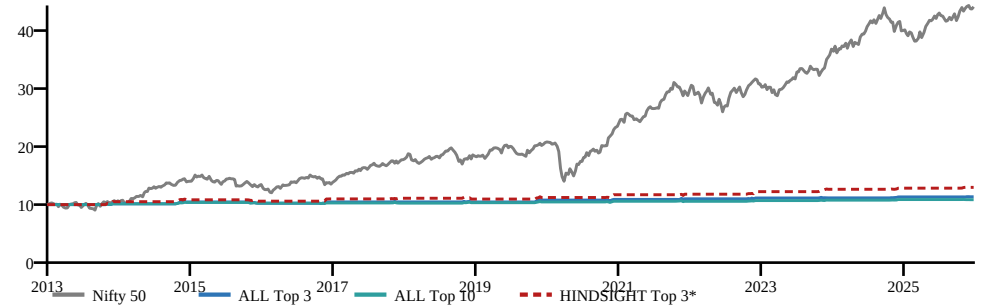
How it is run: each year, for each of the 30 dates (a "sleeve"), the Top-N stocks and their "buy N days before / sell M days after" rule are picked from earlier years; equity is split equally across the sleeves in use and each sleeve across its N stocks; buy at the close of T-n, sell at the close of T+m (T = first trading day on/after the date); cost 0.15% per position; daily mark-to-market; risk-free 6.5%. Policies: **ALL** = all 30 dates; **EXPAND** = dates qualify from their record in earlier years (honest, 2018-2025); **HINDSIGHT*** = dates chosen using the whole period (biased, reference only).

1. Headline measures

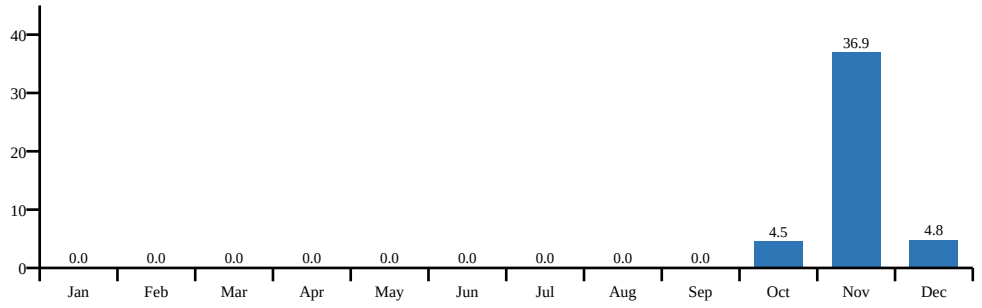
Measure	ALL Top 1 2013-25	EXPAND Top 1 2018-25	HIND.* Top 1 2013-25	ALL Top 3 2013-25	EXPAND Top 3 2018-25	HIND.* Top 3 2013-25	ALL Top 5 2013-25	EXPAND Top 5 2018-25	HIND.* Top 5 2013-25	ALL Top 10 2013-25	EXPAND Top 10 2018-25	HIND.* Top 10 2013-25
CAGR % (idle cash 0%)	1.04	-1.00	2.00	0.98	0.99	2.02	0.72	0.39	1.58	0.63	0.88	1.39
CAGR % (idle cash 6.5%)	7.42	5.25	8.42	7.36	7.38	8.46	7.09	6.73	7.99	6.99	7.26	7.80
Nifty 50 CAGR %, same window	12.12	12.03	12.12	12.12	12.03	12.12	12.12	12.03	12.12	12.12	12.03	12.12
Average fund utilised %	3.89	3.89	4.11	3.73	3.71	3.80	3.73	3.86	3.95	3.74	3.73	3.71
Peak fund utilised %	65	94	82	54	75	66	52	63	63	53	77	76
Days in market %	12	10	11	13	12	13	14	13	13	14	13	14
Return on deployed capital %/yr	27.2	-27.2	50.3	26.5	27.5	53.6	19.6	10.5	39.6	17.0	24.0	37.7
Sharpe (idle cash 0%)	-2.16	-2.01	-1.31	-3.09	-2.25	-1.84	-3.81	-3.09	-2.18	-4.59	-3.31	-2.77
Sharpe (idle cash 6.5%)	0.33	-0.33	0.54	0.43	0.33	0.76	0.34	0.09	0.62	0.32	0.39	0.65
Max drawdown % (idle 0%)	-6.4	-13.8	-5.2	-4.0	-4.8	-4.9	-3.6	-4.5	-3.6	-3.0	-2.5	-5.0
Annual volatility % (idle 0%)	2.51	3.70	3.37	1.78	2.43	2.41	1.51	1.97	2.24	1.28	1.69	1.83
Beta vs Nifty	0.006	0.001	0.015	0.009	0.011	0.019	0.010	0.012	0.020	0.010	0.013	0.016
Trade win %	55.4	53.8	72.3	52.2	54.6	61.0	50.9	49.7	56.5	51.3	54.6	55.5
Profit factor	1.56	0.65	2.54	1.52	1.51	2.41	1.37	1.19	1.95	1.32	1.51	1.86
Worst year %	-1.93	-5.93	-1.57	-2.08	-3.88	-2.17	-2.07	-1.77	-1.66	-1.23	-0.34	-1.83

* HINDSIGHT (red) picks the dates using the whole 2013-2025 results - biased, shown for reference only. EXPAND covers 2018-2025. Sharpe with idle cash at 0% is negative because the account earns less than the 6.5% risk-free rate.

Account value, Rs lakh (idle cash 0%): Nifty 50 vs ALL Top 3 / Top 10 vs HINDSIGHT Top 3 (biased)



Average % of the account deployed, by calendar month (ALL 30 dates, Top 3)



2. Risk and market sensitivity (ALL 30 dates, 2013-2025)

Measure	Top 1	Top 3	Top 5	Top 10	Nifty 50
Annual volatility % (idle 0%)	2.51	1.78	1.51	1.28	16.14
Max drawdown % (idle 0%)	-6.4	-4.0	-3.6	-3.0	-38.4
Longest time below a peak (days)	733	1064	1072	1802	-
Sortino (idle cash 6.5%)	0.52	0.68	0.50	0.47	-
Calmar (idle cash 6.5%)	1.40	3.11	3.24	3.71	-
Daily VaR 95% (invested days) %	0.69	0.47	0.38	0.31	-
Daily CVaR 95% (invested days) %	0.99	0.64	0.57	0.45	-
Worst day %	-1.43	-1.04	-1.34	-1.27	-
Worst month %	-2.25	-1.15	-1.21	-0.72	-
Beta vs Nifty	0.006	0.009	0.010	0.010	-
Alpha % / yr	1.01	0.89	0.62	0.52	-
Correlation with Nifty	0.041	0.084	0.107	0.128	-
Up-market capture	0.02	0.02	0.02	0.02	-
Down-market capture	0.01	0.01	0.01	0.01	-
Information ratio	-0.73	-0.75	-0.76	-0.77	-
Positive years %	69	85	85	69	-

3. Calendar-year returns of the account (idle cash 0%), %

Year	Nifty 50	ALL Top 1	ALL Top 3	ALL Top 5	ALL Top 10
2013	+6.76	+1.12	+1.35	+1.49	+1.35
2014	+31.39	+1.84	+3.18	+3.29	+2.30
2015	-4.06	-1.93	-2.08	-2.07	-1.23
2016	+3.01	+3.24	+2.13	+1.35	+0.40
2017	+28.65	-0.39	+0.05	-0.25	-0.28
2018	+3.15	+1.32	-0.09	+0.24	+0.74
2019	+12.02	+4.83	+3.15	+1.85	+1.25
2020	+14.90	-0.48	+1.05	+0.78	+1.13
2021	+24.12	+0.87	+0.93	+0.42	-0.08
2022	+4.33	+0.17	+1.29	+0.65	+1.18
2023	+20.03	-0.16	+0.16	+0.71	+0.70
2024	+8.80	+3.00	+1.50	+1.00	+0.96
2025	+10.51	+0.31	+0.21	+0.03	-0.23

4. Does picking the "best" dates help? Like-for-like 2018-2025

Portfolio	ALL dates: CAGR % / Sharpe* / worst yr %	EXPANDING (honest)	HINDSIGHT (biased)	EXPAND - ALL, CAGR pts
Top 1	1.22 / 0.37 / -0.48	-1.00 / -0.33 / -5.93	2.65 / 0.79 / 0.45	-2.22
Top 3	1.02 / 0.47 / -0.09	0.99 / 0.33 / -3.88	1.97 / 0.79 / -1.25	-0.03
Top 5	0.71 / 0.36 / 0.03	0.39 / 0.09 / -1.77	1.01 / 0.40 / -1.50	-0.32
Top 10	0.71 / 0.40 / -0.23	0.88 / 0.39 / -0.34	1.57 / 0.89 / 0.54	+0.17

*Sharpe with idle cash earning 6.5%. The honest date filter does not clearly beat trading all dates; only hindsight improves it.

5. Fund utilisation and scaled positions (ALL 30 dates, 2013-2025)

Measure	Top 1	Top 3	Top 5	Top 10
Average fund utilised % (all days)	3.89	3.73	3.73	3.74
Peak fund utilised %	65	54	52	53
Days in market %	12.1	13.1	13.6	14.4
Return on deployed capital %/yr	27.2	26.5	19.6	17.0
Futures scenario: return on margin %/yr (20% margin)	136	133	98	85
Scaled to ~98% peak: CAGR % (idle 0% / 6.5%)	1.57 / 7.84	1.75 / 7.98	1.35 / 7.53	1.14 / 7.32
Scaled: max drawdown %	-9.6	-7.0	-6.8	-5.4

6. Cost sensitivity: CAGR % with idle cash at 6.5% (ALL dates, 2013-2025)

Cost per position	Top 1	Top 3	Top 5	Top 10
0.15% (base)	7.42	7.36	7.09	6.99
0.30%	7.26	7.20	6.93	6.83
0.50%	7.04	6.99	6.72	6.61

7. What the measures mean

Measure	Meaning
CAGR	Compound annual growth: (ending/starting capital)^(1/years) - 1; idle cash earning 0% or 6.5%.
Fund utilised %	Capital in open positions / equity, averaged over ALL days (idle days = 0). Peak = busiest day.
Return on deployed capital	Total P&L / average rupees deployed, per year: what the money earned while working.
Sharpe / Sortino / Calmar	Daily excess return over 6.5%, annualised, / volatility, / downside volatility, / CAGR over |max drawdown|.
VaR / CVaR 95%	Daily loss beaten on 5% of invested days / average loss on those days.
Beta, alpha, capture, IR	Daily regression vs Nifty 50; up/down capture = average return on Nifty up/down days relative to Nifty; IR = excess return per unit of tracking error.
Profit factor	Total gains / total losses in rupees (all trades).
Bootstrap CAGR	Yearly returns resampled 5,000 times: 5th-95th percentile of CAGR.

8. Limits

13 test years (8 for the honest date filter); the 30 dates overlap in time. Survivorship bias: stock list = today's F&O universe. Cash-settled P&L at the close, flat 0.15% cost, no slippage, margin interest or tax; SHORT trades assumed possible at the same cost (they need futures); the futures return-on-margin line is a rough scenario only. HINDSIGHT rows use future information and are not achievable. Every figure in this report is traceable in **November_Portfolio_Report_Data.xlsx** (sheets Report_Numbers, Verification, All_Measures).